



An Empirical Study on Configuration Robustness of Unsupervised IDS

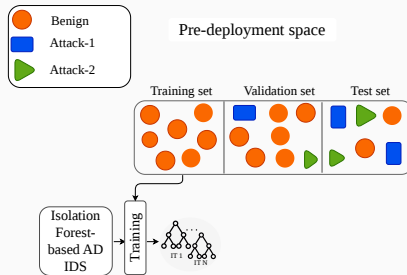
Omar Anser¹, Jerome Francois^{1,3}, Isabelle Chrisment^{2,3}

¹SnT - University of Luxembourg ²Universite de Lorraine, CNRS, LORIA ³Inria

NOMS 2026

ML-based AD IDS Workflow: Train $\rightarrow$ Validate $\rightarrow$ Test $\rightarrow$ Deploy

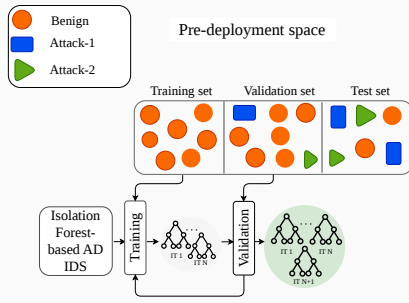
- The model is trained on a training set $\mathcal{D}_{\text{train}}$.



Deployment space

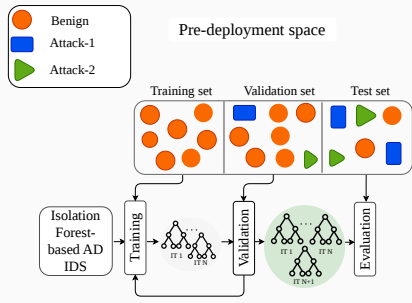
ML-based AD IDS Workflow: Train $\rightarrow$ Validate $\rightarrow$ Test $\rightarrow$ Deploy

- The model is trained on a training set $\mathcal{D}_{\text{train}}$.
- A separate validation split $\mathcal{D}_{\text{val}}$ to choose the model hyperparameters θ .



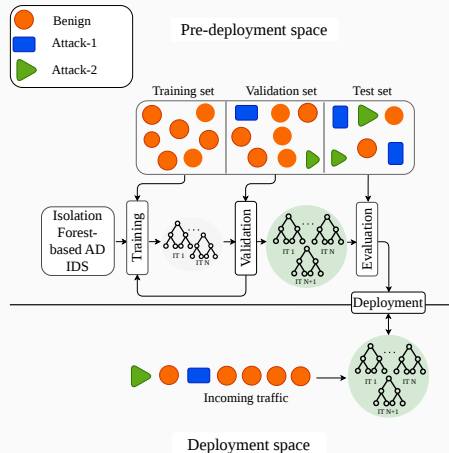
ML-based AD IDS Workflow: Train $\rightarrow$ Validate $\rightarrow$ Test $\rightarrow$ Deploy

- The model is trained on a training set $\mathcal{D}_{\text{train}}$.
- A separate validation split $\mathcal{D}_{\text{val}}$ to choose the model hyperparameters θ .
- A final test set, $\mathcal{D}_{\text{test}}$, measures performance on unseen data.



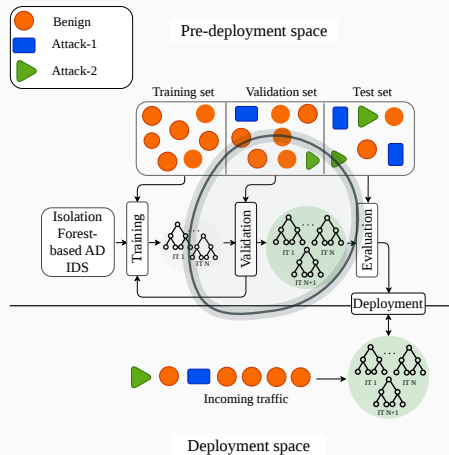
ML-based AD IDS Workflow: Train $\rightarrow$ Validate $\rightarrow$ Test $\rightarrow$ Deploy

- The model is trained on a training set $\mathcal{D}_{\text{train}}$.
- A separate validation split $\mathcal{D}_{\text{val}}$ to choose the model hyperparameters θ .
- A final test set, $\mathcal{D}_{\text{test}}$, measures performance on unseen data.
- The selected model is deployed to monitor live traffic.



ML-based AD IDS Workflow: Train $\rightarrow$ Validate $\rightarrow$ Test $\rightarrow$ Deploy

- The model is trained on a training set $\mathcal{D}_{\text{train}}$.
- A separate validation split $\mathcal{D}_{\text{val}}$ to choose the model hyperparameters θ .
- A final test set, $\mathcal{D}_{\text{test}}$, measures performance on unseen data.
- The selected model is deployed to monitor live traffic.



$\Rightarrow$ Focus on the impact of the **configuration of hyperparameters (HPs)** on **AD IDS performance**.

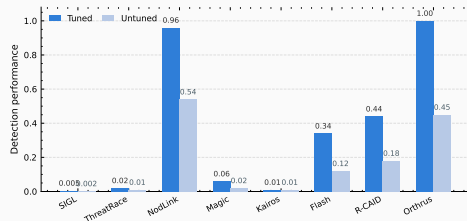
Related Work and Gap

Broader Machine Learning

- **Hutter'14 / van Rijn'18^{1,2}**: a few HPs explain most performance variation.
- **DOMAINBED / WILDS / HYPERTIME³⁻⁵**: HPs optimized on one data distribution may fail after the distribution changes.

Security / IDS

- **Arp'22 / Pendlebury'19 / Masum'21⁶⁻⁸**: IDS results depend strongly on tuning and validation.
- **Bilot'25⁹**: 8 provenance-graph IDSs; tuned vs. untuned changes the conclusion.
- Static analyses of configuration impact.



Source: Bilot et al., USENIX Security 2025

⇒ **Gap:** no direct quantification for AD IDS under network context changes.

[1] Hutter et al., "An Efficient Approach for Assessing Hyperparameter Importance", ICML 2014; [2] van Rijn and Hutter, "Hyperparameter Importance Across Datasets", KDD 2018.

[3] Gulrajani and Lopez-Paz, "In Search of Lost Domain Generalization", arXiv 2020; [4] Koh et al., "WILDS: A Benchmark of in-the-Wild Distribution Shifts", ICML 2021.

[5] Zhang et al., "HyperTime: Hyperparameter Optimization for Combating Temporal Distribution Shifts", ACM MM 2024; [6] Arp et al., "Dos and Don'ts of Machine Learning in Computer Security", USENIX Security 2022.

[7] Pendlebury et al., "TESSERACT: Eliminating Experimental Bias in Malware Classification across Space and Time", USENIX Security 2019; [8] Masum et al., "Bayesian Hyperparameter Optimization for Deep Neural Network-Based Network Intrusion Detection", IEEE BigData 2021.

[9] Bilot et al., "Sometimes Simpler is Better: A Comprehensive Analysis of State-of-the-Art Provenance-Based Intrusion Detection Systems", USENIX Security 2025.

Operational Setting and Research Questions

Operational setting

- Configuring θ can be costly $\rightarrow$ often fixed once.
- Network context changes over time: traffic volume, composition, topology, and attack types evolve.

Research questions

1. If θ is reused, how much performance is lost compared with reconfiguring for the current network context?
2. Can simple low-budget strategies approach the performance of full reconfiguration?

Experimental Setup: Benchmark Context

- Corrected IDS2017¹ / IDS2018² provide day-level traffic datasets $D_0, \dots, D_{13}$.

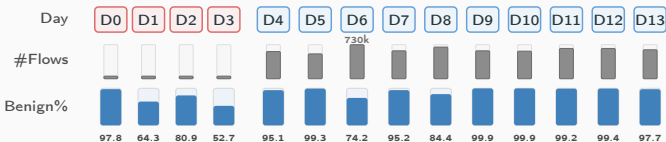
Day D0 D1 D2 D3 D4 D5 D6 D7 D8 D9 D10 D11 D12 D13

¹ Sharafaldin et al., "Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization", ICISSP 2018.

² CSE-CIC-IDS2018 on AWS, CSE and CIC, 2018; official dataset page recommends citing Sharafaldin et al., ICISSP 2018.

Experimental Setup: Benchmark Context

- Corrected IDS2017¹ / IDS2018² provide day-level traffic datasets $D_0, \dots, D_{13}$.
- Flow volume varies; benign traffic is above 95% on 8/14 days, and varies on the others.

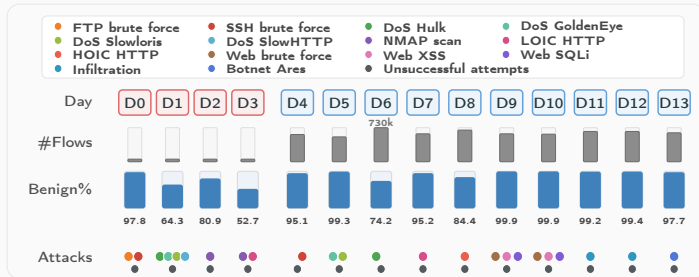


¹ Sharafaldin et al., "Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization", ICISPP 2018.

² CSE-CIC-IDS2018 on AWS, CSE and CIC, 2018; official dataset page recommends citing Sharafaldin et al., ICISPP 2018.

Experimental Setup: Benchmark Context

- Corrected IDS2017¹ / IDS2018² provide day-level traffic datasets $D_0, \dots, D_{13}$.
- Flow volume varies; benign traffic is above 95% on 8/14 days, and varies on the others.
- Attack types also vary by day.

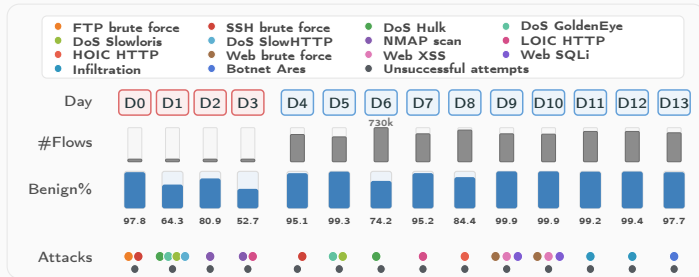


¹ Sharafaldin et al., "Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization", ICISSP 2018.

² CSE-CIC-IDS2018 on AWS, CSE and CIC, 2018; official dataset page recommends citing Sharafaldin et al., ICISSP 2018.

Experimental Setup: Benchmark Context

- Corrected IDS2017¹ / IDS2018² provide day-level traffic datasets $D_0, \dots, D_{13}$.
- Flow volume varies; benign traffic is above 95% on 8/14 days, and varies on the others.
- Attack types also vary by day.

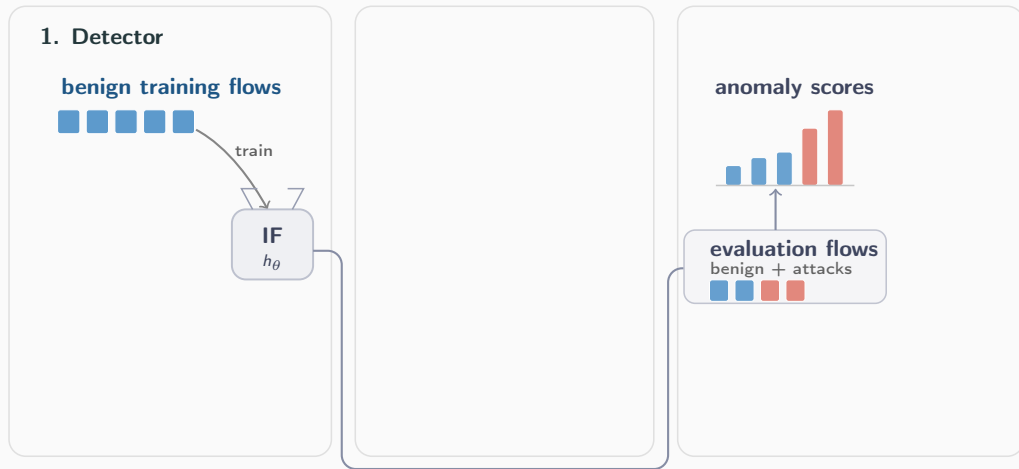


⇒ Each day defines a distinct dataset and network context; moving across days represents a network context change.

¹ Sharafaldin et al., "Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization", ICISSP 2018.

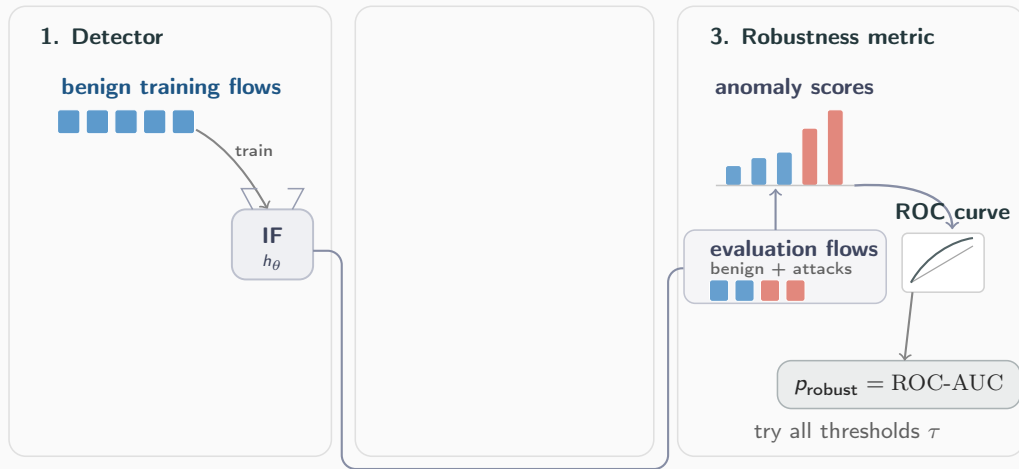
² CSE-CIC-IDS2018 on AWS, CSE and CIC, 2018; official dataset page recommends citing Sharafaldin et al., ICISSP 2018.

Experimental Setup: IF, BO, and Robustness Metric



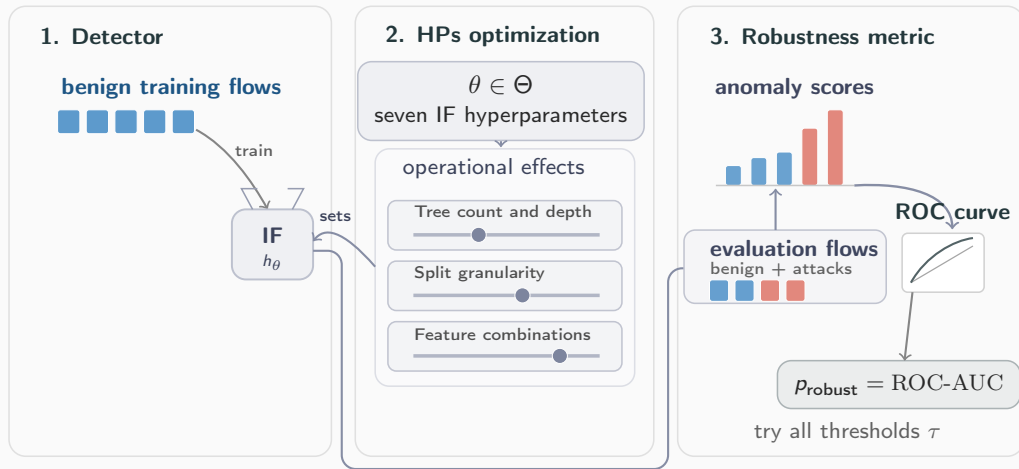
⇒ Isolation Forest (IF): train on benign flows, score benign + attack flows.
Lightweight enough for many configuration runs.

Experimental Setup: IF, BO, and Robustness Metric



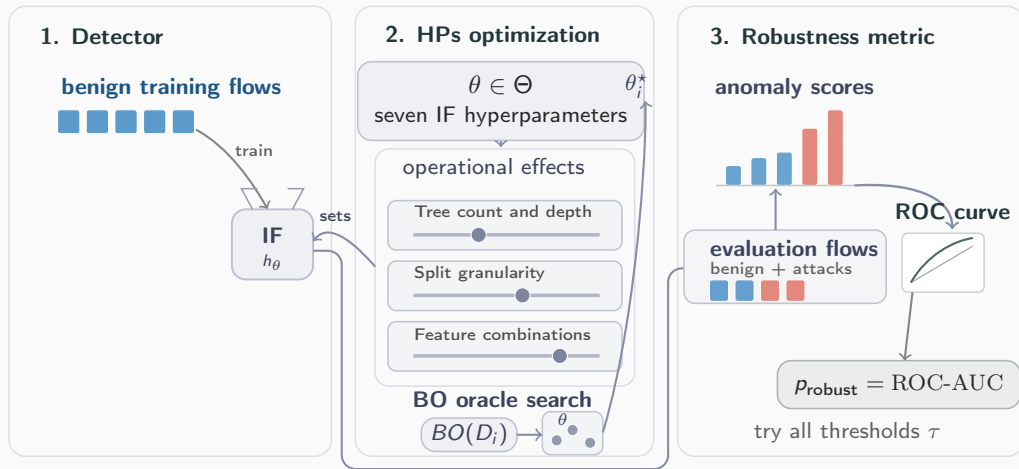
$\Rightarrow$ ROC-AUC: try all alert thresholds; area under the curve = p_{robust} .
Higher means better attack/benign separation.

Experimental Setup: IF, BO, and Robustness Metric



$\Rightarrow \theta$ groups seven IF hyperparameters by their operational effect.

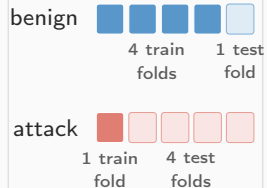
Experimental Setup: IF, BO, and Robustness Metric



$\Rightarrow$ Bayesian Optimization (BO) gives the oracle reference: $\theta_i^* = BO(D_i)$.

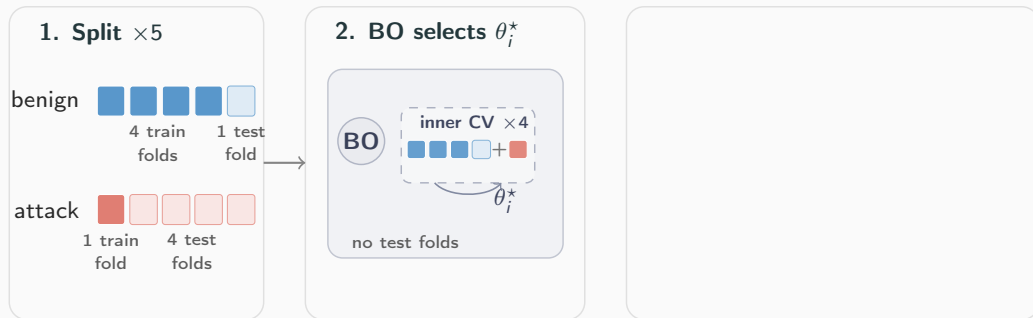
Experimental Setup: Data Splits and BO Search

1. Split $\times 5$



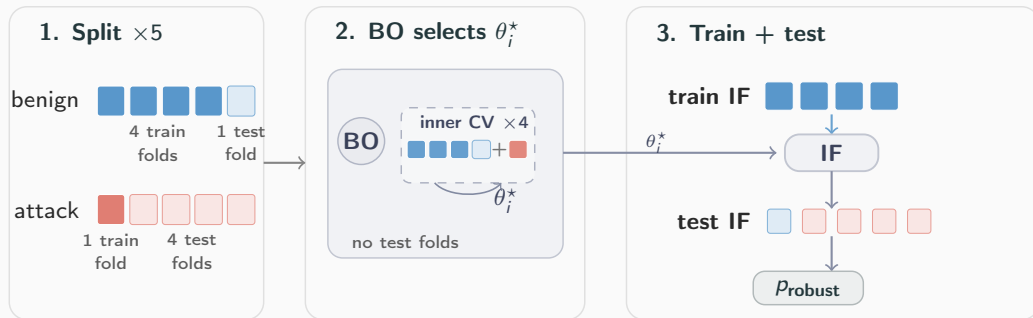
$\Rightarrow$ Outer folds define D_{train} and D_{test} without mixing test traffic into configuration.

Experimental Setup: Data Splits and BO Search



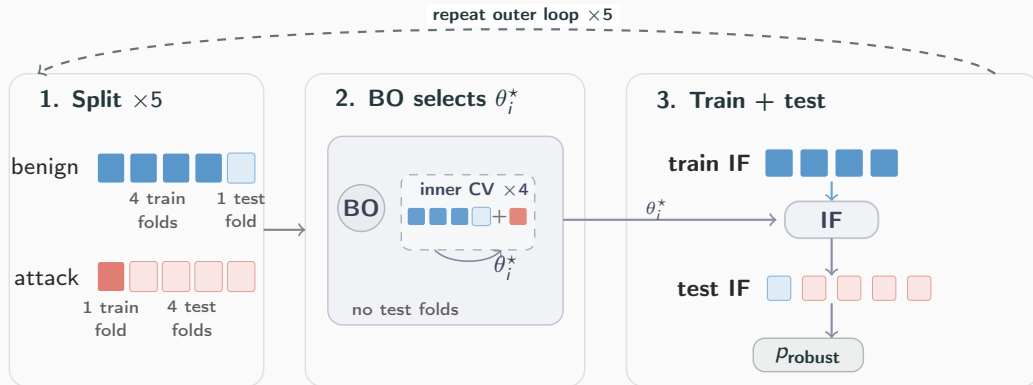
$\Rightarrow$ BO uses only training-side data: inner folds form D_{val} to select θ_i^* .

Experimental Setup: Data Splits and BO Search



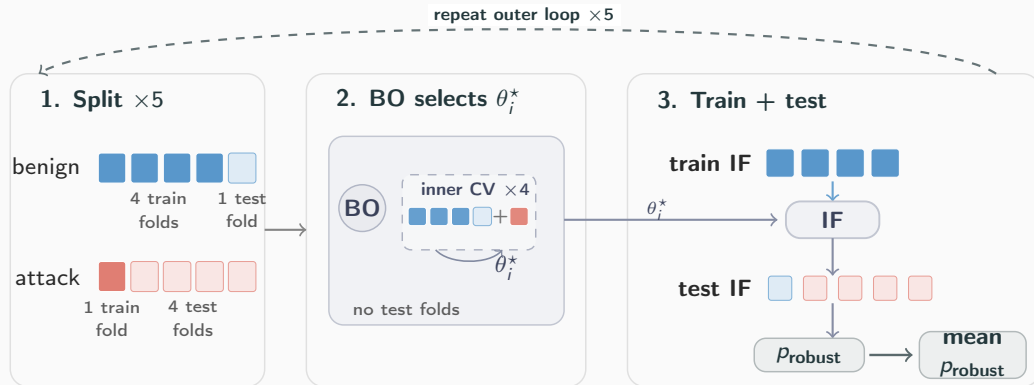
$\Rightarrow$ Evaluation trains IF on benign training folds and tests on held-out benign + attack folds.

Experimental Setup: Data Splits and BO Search



$\Rightarrow$ The same outer construction is repeated over the five folds.

Experimental Setup: Data Splits and BO Search



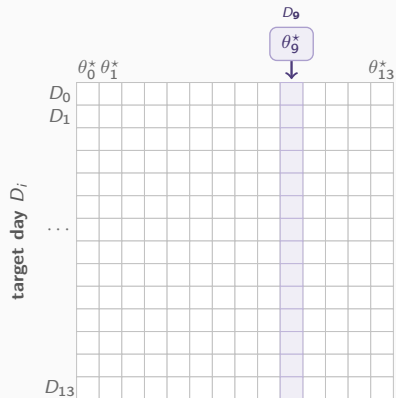
$\Rightarrow$ The reported oracle reference is mean p_{robust} .

Research Question 1

If θ is reused, how much performance is lost compared with reconfiguring for the current network context?

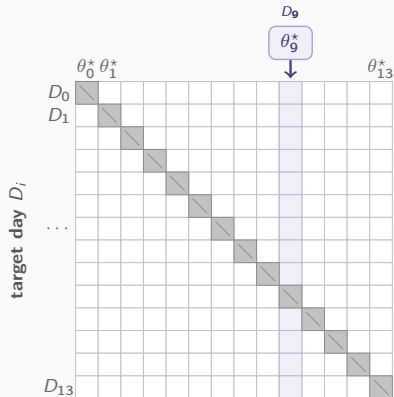
Day-to-Day Transferability: Method

- BO gives one oracle configuration per day; pick a source D_9 with θ_9^* .



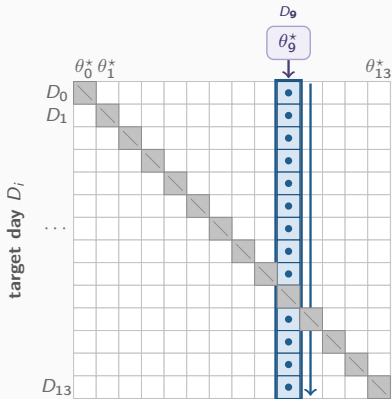
Day-to-Day Transferability: Method

- BO gives one oracle configuration per day; pick a source D_9 with θ_9^* .
- Diagonal cells are oracle references, not transfer evaluations.



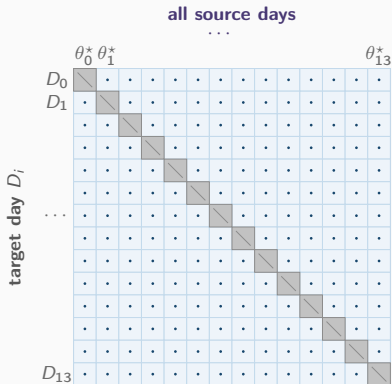
Day-to-Day Transferability: Method

- BO gives one oracle configuration per day; pick a source D_9 with θ_9^* .
- Diagonal cells are oracle references, not transfer evaluations.
- Keep θ_9^* fixed and evaluate it on every target day D_i , excluding $i = 9$.



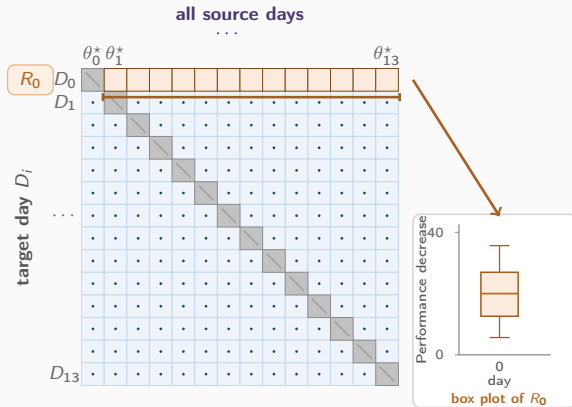
Day-to-Day Transferability: Method

- BO gives one oracle configuration per day; pick a source D_9 with θ_9^* .
- Diagonal cells are oracle references, not transfer evaluations.
- Keep θ_9^* fixed and evaluate it on every target day D_i , excluding $i = 9$.
- Repeating this for all source days fills the off-diagonal matrix.



Day-to-Day Transferability: Method

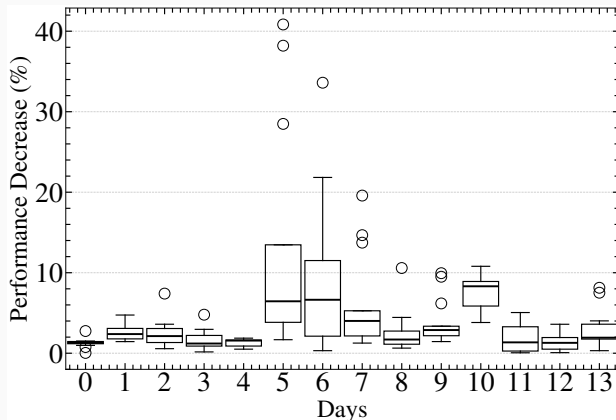
- BO gives one oracle configuration per day; pick a source D_9 with θ_9^* .
- Diagonal cells are oracle references, not transfer evaluations.
- Keep θ_9^* fixed and evaluate it on every target day D_i , excluding $i = 9$.
- Repeating this for all source days fills the off-diagonal matrix.
- Each row R_i groups the transfer losses for one target day D_i .



⇒ Lower row losses indicate safer configuration reuse.

Day-to-Day Transferability: Results

- Each box summarizes one target day
 $R_i = \{\Delta(i, j) : j \neq i\}$.
- Wide boxes and outliers show sensitivity to the chosen source day.
- Days 5–7 and 10 are the riskiest; losses can approach 40%.

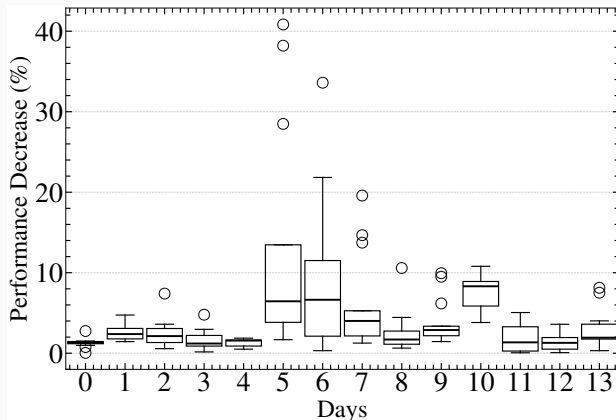


Transfer loss distributions by target day

Day-to-Day Transferability: Results

- Each box summarizes one target day
 $R_i = \{\Delta(i, j) : j \neq i\}$.
- Wide boxes and outliers show sensitivity to the chosen source day.
- Days 5–7 and 10 are the riskiest; losses can approach 40%.

Cross-year transferability: transfer IDS2017-tuned configurations to IDS2018 for common attack types; same behavior.

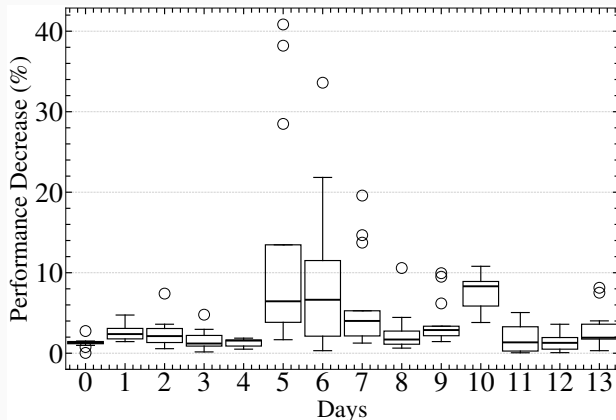


Transfer loss distributions by target day

Day-to-Day Transferability: Results

- Each box summarizes one target day
 $R_i = \{\Delta(i, j) : j \neq i\}$.
- Wide boxes and outliers show sensitivity to the chosen source day.
- Days 5–7 and 10 are the riskiest; losses can approach 40%.

Cross-year transferability: transfer IDS2017-tuned configurations to IDS2018 for common attack types; same behavior.



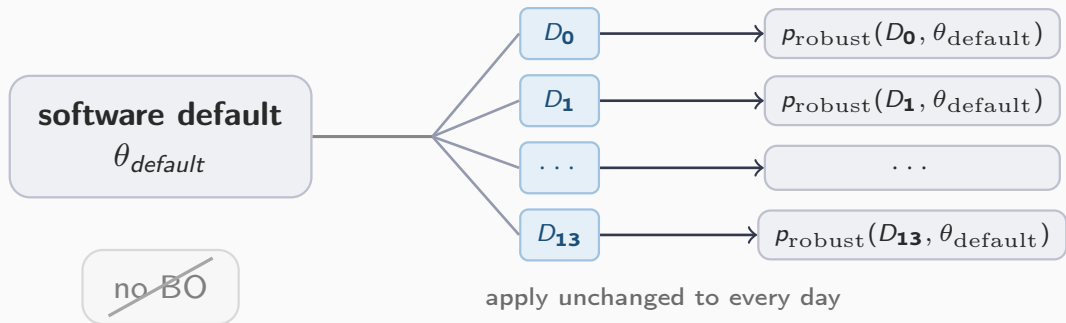
Transfer loss distributions by target day

⇒ **RQ1:** The cost of reusing θ depends on the target context: some days lose up to 40%; same-attack reuse may still need reconfiguration.

Research Question 2

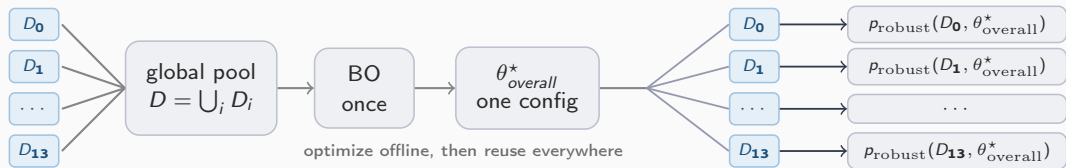
Can simple low-budget strategies approach the performance of full reconfiguration?

Alternative 1 – Default



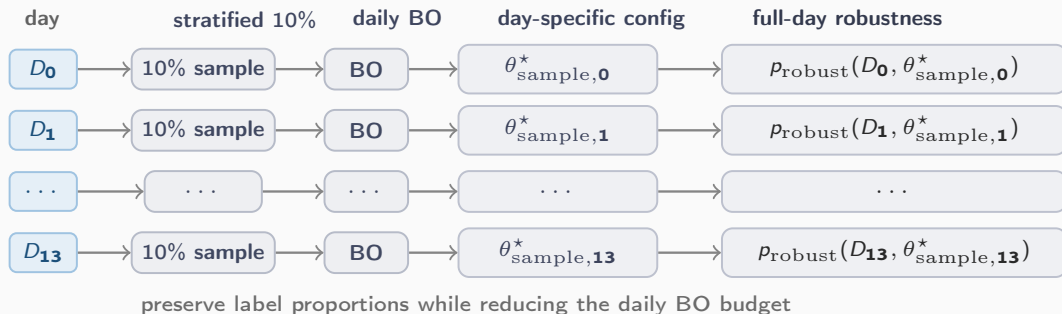
$\Rightarrow$ Zero tuning cost; tests whether the library default is robust enough across changing network contexts.

Alternative 2 – Single Overall



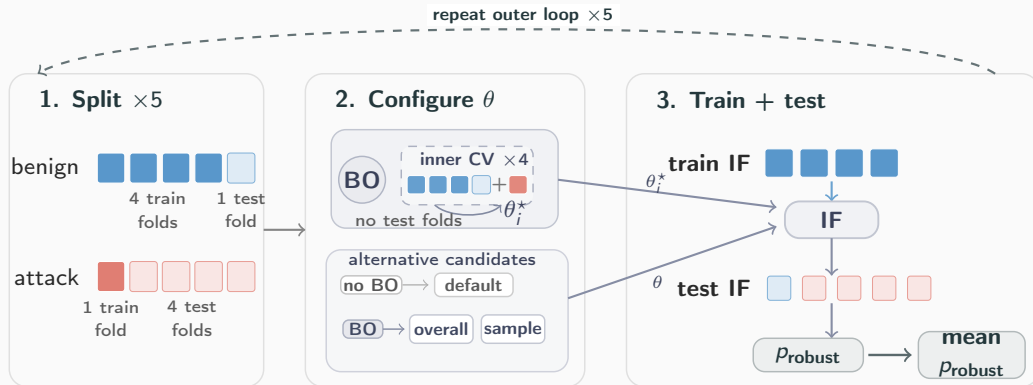
$\Rightarrow$ Optimize once on heterogeneous historical traffic; amortize BO effort into one reusable configuration.

Alternative 3 – 10% Sample BO



⇒ Keep day-specific adaptation, but reduce BO cost by tuning on a stratified 10% subset.

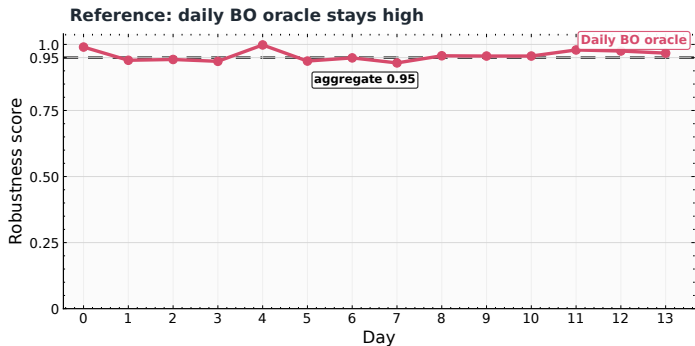
Nested Cross-Validation



$\Rightarrow$ Default uses no BO; overall and sample are BO-derived alternative candidates, then all use the same IF train/test evaluation.

Main Results

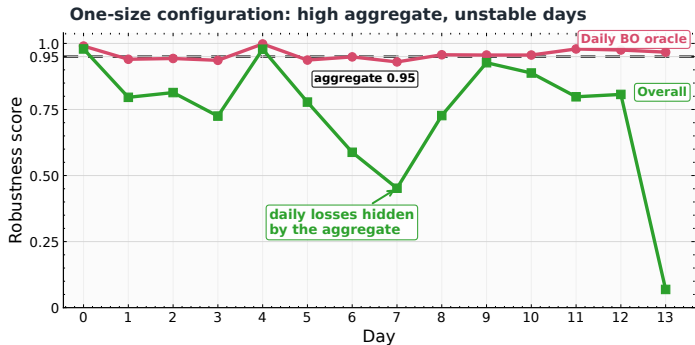
- The daily BO oracle remains consistently high.



Daily robustness across configuration strategies

Main Results

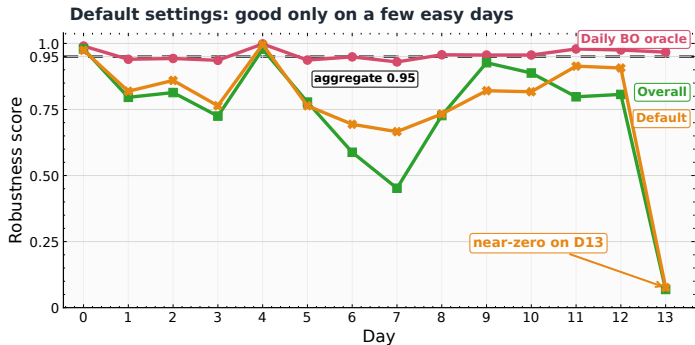
- The daily BO oracle remains consistently high.
- The aggregate score hides unstable days.



Daily robustness across configuration strategies

Main Results

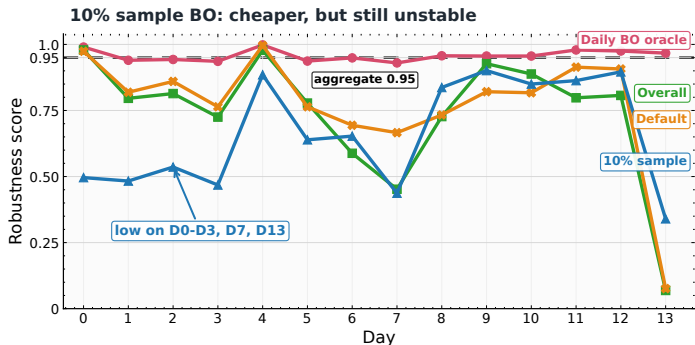
- The daily BO oracle remains consistently high.
- The aggregate score hides unstable days.
- Fixed configurations are not uniformly robust.



Daily robustness across configuration strategies

Main Results

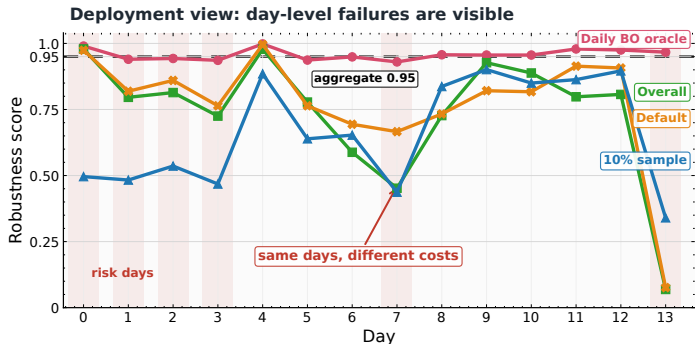
- The daily BO oracle remains consistently high.
- The aggregate score hides unstable days.
- Fixed configurations are not uniformly robust.
- Under-sampling lowers tuning effort, but not instability.



Daily robustness across configuration strategies

Main Results

- The daily BO oracle remains consistently high.
- The aggregate score hides unstable days.
- Fixed configurations are not uniformly robust.
- Under-sampling lowers tuning effort, but not instability.

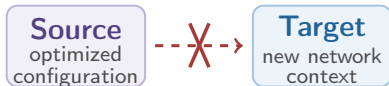


Daily robustness across configuration strategies

⇒ **RQ2:** Simple low-budget strategies do not reliably match full reconfiguration.

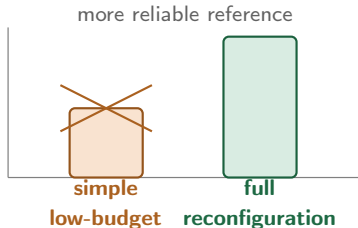
Takeaway

Transfer was unreliable



reuse can lose robustness
up to 40% in risky contexts

Low-budget was unstable



⇒ Reconfigure for each network context
for more reliable AD IDS performance.

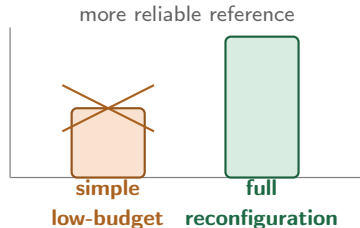
Takeaway

Transfer was unreliable



reuse can lose robustness
up to 40% in risky contexts

Low-budget was unstable



⇒ **Reconfigure for each network context
for more reliable AD IDS performance.**

**Guided
low-budget**

O. Anser, J. Francois, and I. Chrisment, IEEE/IFIP NOMS.
“Auto-tuning of Hyper-parameters for Detecting
Network Intrusions via Meta-learning.”

Acknowledgments

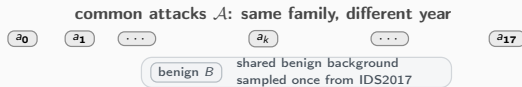
- Partial support from the French National Research Agency (ANR) through the France 2030 initiative: **Superviz**, ANR-22-PECY-0008.
- Additional support from the Luxembourg National Research Fund (FNR): **C23/IS/18088425/COCTEL**.

The opinions expressed are those of the authors and do not necessarily reflect the views of the French government.

Questions?

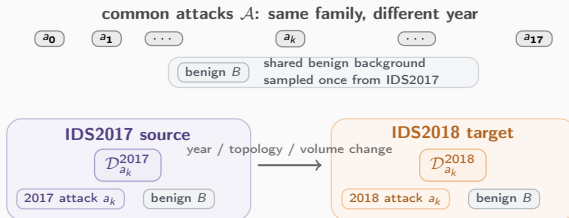
Backup: Cross-Year Attack Transfer Method

- Select a common attack family a_k and shared benign background B .



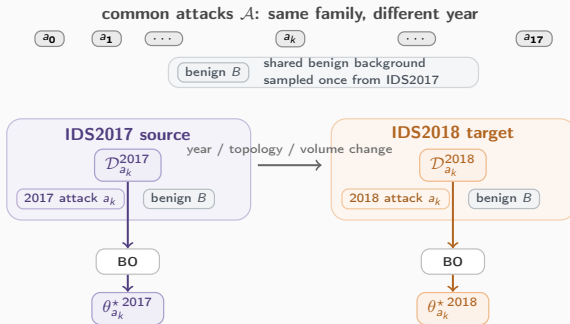
Backup: Cross-Year Attack Transfer Method

- Select a common attack family a_k and shared benign background B .
- Build paired IDS2017/IDS2018 datasets from a_k and B .



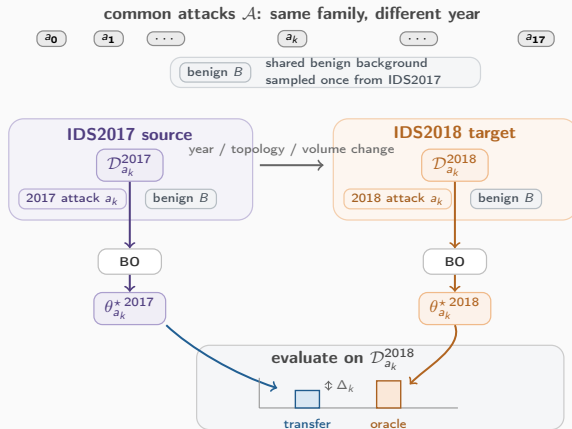
Backup: Cross-Year Attack Transfer Method

- Select a common attack family a_k and shared benign background B .
- Build paired IDS2017/IDS2018 datasets from a_k and B .
- BO gives $\theta_{a_k}^{*2017}$ and the IDS2018 oracle $\theta_{a_k}^{*2018}$.



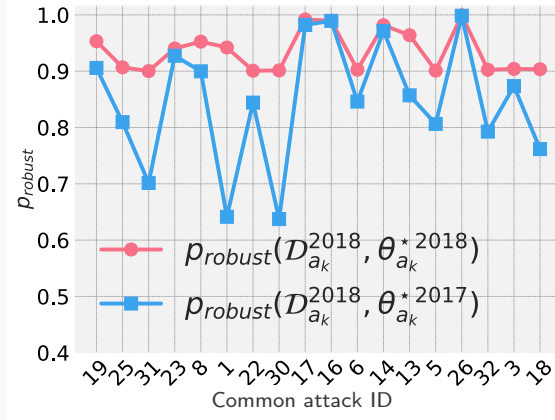
Backup: Cross-Year Attack Transfer Method

- Select a common attack family a_k and shared benign background B .
- Build paired IDS2017/IDS2018 datasets from a_k and B .
- BO gives $\theta_{a_k}^{*2017}$ and the IDS2018 oracle $\theta_{a_k}^{*2018}$.
- Evaluate both configurations on $\mathcal{D}_{a_k}^{2018}$ and measure the gap.



Backup: Cross-Year Attack Transfer Results

- Each attack family compares IDS2017-tuned transfer against the IDS2018 oracle.
- Attacks 23, 17, 16, 14, and 26 transfer close to the oracle.
- Attacks 31, 1, and 30 show much larger cross-year gaps.
- Same attack family is therefore not enough to guarantee robust reuse.



IDS2017 configuration transferred to IDS2018 by attack family

⇒ **RQ1:** Same-attack reuse across years can still require reconfiguration.